

TEST SUMMARY REPORT: StudyFlow E2E & System Testing

Project Name: StudyFlow

Domain: <https://studyflow.ge>

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Environment: Staging / Production

Execution Type: Manual & Automated (API + Cypress E2E + GitHub Actions CI/CD)

1. Executive Summary

This report documents the final quality assessment for the **StudyFlow** web application. Testing covered the complete Software Testing Life Cycle (STLC), including requirements analysis, UI/UX evaluation, manual & API testing, defect tracking, and full-stack E2E test automation integrated into a CI/CD pipeline.

The target system successfully passed all critical business workflows and security verification gates. The automated test suite demonstrated **100% stability** in the GitHub Actions regression pipeline, proving the application is production-ready.

2. Scope of Testing

- **Manual Functional Testing:** Verification of 33 standard Test Cases covering Authentication, Subjects, Assignments, Schedule, Exams, and AI Assistant modules.
- **API Testing:** Validation of 21 backend API endpoints for status codes, payloads, query parameters, and response headers.
- **Defect Management:** Logging, tracking, and verifying 6 reported bugs through their complete lifecycle.
- **Automation Testing:** 14 End-to-End (E2E) workflow test suites implemented in Cypress.

- **CI/CD Integration:** Automated workflow execution on code push/PR using GitHub Actions.

3. Test Execution Summary

Test Category	Planned / Executed	Passed	Failed / Resolved	Pass Rate
Manual Test Cases	33	33	0	100%
API Test Endpoints	21	21	0	100%
Database Tests	5	5	0	100%
Automated Cypress E2E	14	14	0	100%
Total Test Execution	73	73	0	100%

4. Defect Metrics & Resolution Summary

During the manual and API execution phases, a total of **6 defects** were identified, logged in the bug tracking repository, and re-tested upon resolution:

- **Total Bugs Identified:** 6
- **Resolved & Verified:** 6 (100% Fix Rate)
- **Open / Unresolved Bugs:** 0
- **Defect Breakdown:**
 - 2 UI/UX Selector & Modal alignment issues.
 - 2 API Payload validation & required field handling bugs (Exam/Schedule inputs).
 - 1 Browser event handling issue (dialog on logout).window:confirm
 - 1 Non-text element target mismatch (Profile SVG logout button).

5. Automated E2E Workflows Verified (Cypress)

- **User Lifecycle:** Dynamic registration with timestamp credentials, authorization verification, and secure logout.

- **Subject Management (CRUD):** Successful subject creation, UI visibility assertions, and teardown.
- **Assignment Lifecycle:** Assignment creation, dynamic completion status change, numerical grade submission, and deletion.
- **Schedule & Exam Modules:** Slot allocations, date validation checks, and element deletion.
- **AI Assistant:** Real-time chat interaction and prompt/response validation.
- **CI/CD Pipeline:** Fully automated Chrome execution in GitHub Actions with artifact generation enabled.

6. Final Recommendation & Sign-Off

Based on the execution of **33 Test Cases**, **21 API Endpoint tests**, resolution of all **6 defects**, and the successful execution of **14 Cypress E2E Automation tests** in the CI/CD pipeline, the StudyFlow platform has satisfied all quality requirements.

Final Verdict: APPROVED FOR PRODUCTION RELEASE